

Assignment 2

Due Monday 2 February at the beginning of class

This Assignment is based on Chapters 1 and 2 of our textbook,¹ and on the lectures.

DO THE FOLLOWING Exercises from Chapter 2 (see pages 29–31):

- **Exercise 2.1.1** *Note that Saxe defines open and closed sets on page 16. Please use those definitions.*
- **Exercise 2.1.4**
- **Exercise 2.1.6**
- **Exercise 2.1.7** *State your claim, and then prove it.*
- **Exercise 2.1.10**
- **Exercise 2.1.13** *Do parts (a) and (b) only.*
- **Exercise 2.2.2** *You may use, without proof, the fact that a countable union of countable or finite sets is countable.*
- **Exercise 2.3.1**
- **Exercise 2.3.5** *To show that the formula defines a norm, you may use, without proof, the triangle inequality for integrals, that is, $|\int_E f(t) dt| \leq \int_E |f(t)| dt$.*
- **Exercise 2.3.8** *You may use, without proof, the fact that $\mathbb{R}$ is complete.*

DO THE FOLLOWING ADDITIONAL PROBLEMS.

P3. *In lecture² I solved a classic boundary value problem (BVP) by writing $u(x)$ as an integral of the data $\phi(x)$. This is nearly the same, or very slightly easier. In fact $\phi \in L^1(0, 1)$ suffices, but we need not worry about generality for now.*

Consider the following BVP, with data $\phi \in C([0, 1])$:

$$\begin{aligned} -u''(x) &= \phi(x) && \text{for all } x \in (0, 1) \\ u(0) &= 0 \\ u'(1) &= 0 \end{aligned}$$

Solve this problem by integration. In particular, write the solution as

$$u(x) = \int_0^1 k(x, t) \phi(t) dt,$$

and give the precise formula for $k(x, t)$. Is k a continuous function of its two variables? Is it bounded?

¹K. Saxe, *Beginning Functional Analysis*, Springer 2010.

²See these slides: <https://bueler.github.io/fa/assets/slides/S26/week2.pdf>.